Exploitation de l'EGT 2010

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Préparation du traitement

Installation des librairies et importation des données

```
library(tidyverse)
library(haven)
library(viridis)
library(questionr)

# Viridis colors as default for ggplot2 : easier to read by those with colorblindness, and print well i
options(ggplot2.continuous.colour="viridis")
options(ggplot2.continuous.fill = "viridis")
scale_colour_discrete <- scale_colour_viridis_d
scale_fill_discrete <- scale_fill_viridis_d

menages <- read_sas("../data/men2010.sas7bdat")
individus <- read_sas("../data/ind2010.sas7bdat")
deplacements <- read_sas("../data/dplct2010.sas7bdat")
trajets <- read_sas("../data/trajet2010.sas7bdat")

# Recode to factors, except a few columns
menages <- menages %>%
  mutate_at(vars(-POIDSM, -MNP, -MNP5, -MNPMOB, -MNPACT, -LOY_HC, -CHG, -LOY_PARK, -NBPI, -SURF, -ANEM,
individus <- individus %>%
  mutate_at(vars(-POIDSP, -AGE, -NBDEPL, -NBDEPLVP, -NBDEPLVPC, -NBDEPLTC, -NBDEPLVELO, -NBDEPL2RM, -NBI
deplacements <- deplacements %>%
  mutate_at(vars(-POIDSP, -DPORTEE, -DUREE, -NBTRAJ, -NBTRAJVP, -NBTRAJVPC, -NBTRAJTC, -NBTRAJVELO, -NBT
trajets <- trajets %>%
  mutate_at(vars(-POIDSP, -TPORTEE), factor)
```

```
CAT_des_CS8_non_renseignee <- CS8_non_renseignee %>%
  filter(CS8 == "undefined", AGE >= 15) %>%
  group_by(CAT) %>%
  count
```

Table 1: 15+ ans dont la CS8 n'est pas renseignée, par catégorie de personne

CAT	n
3	2
4	3
5	83
6	3

CAT	n
7	7
8	3
9	159
14	2
18	2
19	4
20	94
21	4
22	107
23	82
24	1003
28	1

```
CAT_simplifiee_des_CS8_non_renseignee <- CS8_non_renseignee %>%
  filter(CS8 == "undefined", AGE >= 15) %>%
  mutate(CAT = if_else(CAT %in% c(10:15, 25:30), "actifs occupes", "autres")) %>%
  group_by(CAT) %>%
  count
```

Table 2: 15+ ans dont la CS8 n'est pas renseignée, répartition entre actifs occupés et les autres

CAT	n
actifs occupes	3
autres	1556

Distribution des temps de déplacements par jour et par personne

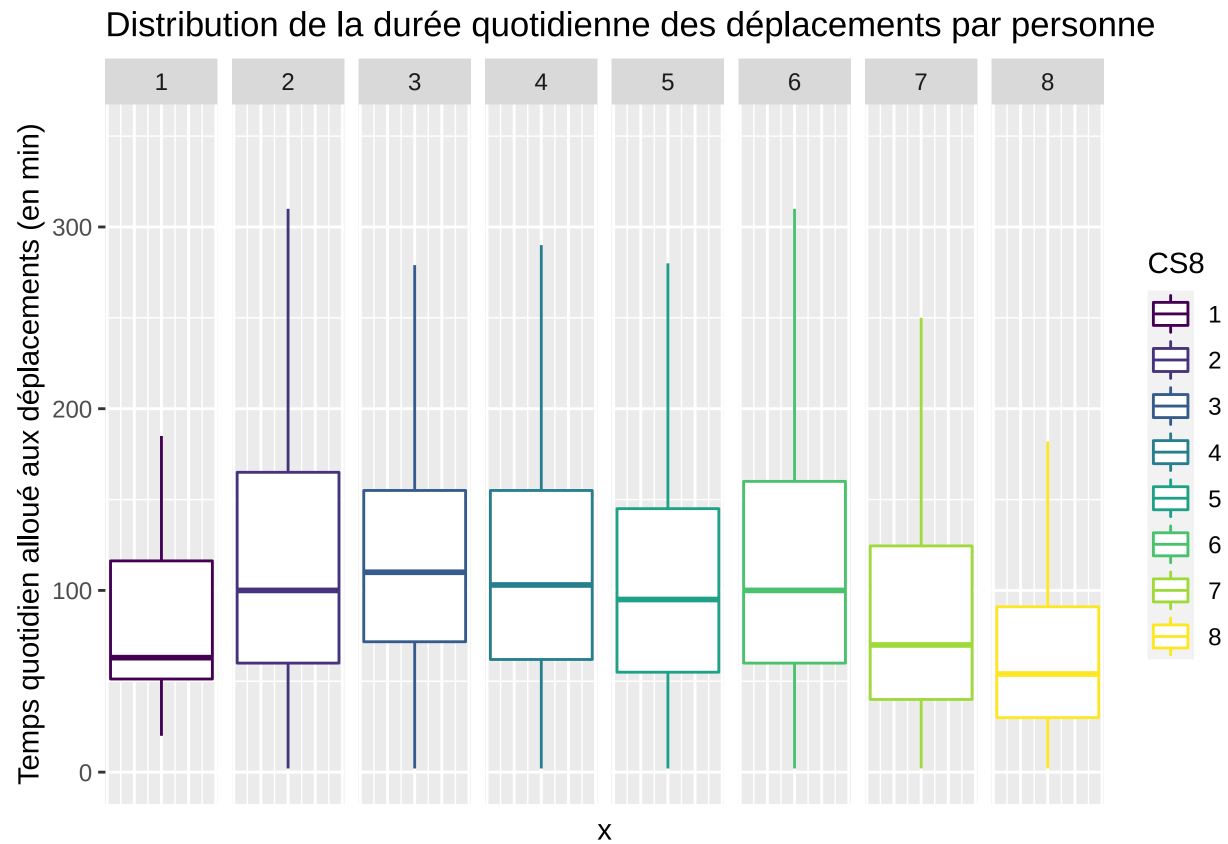
```
temps_quotid_personne <- deplacements %>%
  group_by(NQUEST, NP) %>%
  summarise(temps = sum(DUREE)) %>%
  left_join(individus, by = c("NQUEST", "NP")) %>%
  filter(!is.na(CS8)) %>%
  group_by(CS8)
```

```
## Warning: Column `NQUEST` joining factors with different levels, coercing to
## character vector

## Warning: Column `NP` joining factors with different levels, coercing to
## character vector
```

```
ggplot(temps_quotid_personne, aes(x = 1, y = temps, color = CS8)) +
  facet_grid(cols = vars(CS8)) +
  geom_boxplot(outlier.shape = NA) +
  coord_cartesian(ylim = c(0, 350)) +
  ggtitle("Distribution de la durée quotidienne des déplacements par personne") +
  ylab("Temps quotidien alloué aux déplacements (en min)") +
  theme(axis.text.x = element_blank(), axis.ticks.x = element_blank())
```

```
## Warning: Removed 571 rows containing non-finite values (stat_boxplot).
```



–>

Qui sont les membres de la PCS 56 ?

Lien à la personne de référence

```
ALL_LIENPREF <- individus %>%
  mutate(LIENPREF = fct_collapse(LIENPREF, "pere-mere" = c("1", "2"))) %>%
  mutate(LIENPREF = fct_other(LIENPREF, keep = "pere-mere")) %>%
  group_by(LIENPREF) %>%
  count

PCS56_LIENPREF <- individus %>%
  filter(CS24L == 56) %>%
  mutate(LIENPREF = fct_collapse(LIENPREF, "pere-mere" = c("1", "2"))) %>%
  mutate(LIENPREF = fct_other(LIENPREF, keep = "pere-mere")) %>%
  group_by(LIENPREF) %>%
  count

lienpref <- ALL_LIENPREF %>% full_join(PCS56_LIENPREF, by = c("LIENPREF"), suffix = c(" all", " pcs_56"
```

Table 3: Lien à la personne de référence

LIENPREF	n all	n pcs_56
pere-mere	22894	1023
Other	12281	81

Modèle logistique

Préparation

```r
library(survey)
```

```
## Loading required package: grid

## Loading required package: Matrix

##
## Attaching package: 'Matrix'

## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack

## Loading required package: survival

##
## Attaching package: 'survey'

## The following object is masked from 'package:graphics':
##
##     dotchart
```

```r
library(gtsummary)
library(forestmodel)

prepare_individus_for_regression <- function (data) {
  data <- data %>%
    # Variable "is_PCS56"
    mutate(is_PCS56 = (CS24L == 56) & (!is.na(CS24L))) %>%
    # Adapation des sexes
    mutate(SEXE = fct_recode(SEXE, "Homme" = "1", "Femme" = "2")) %>%
    # 15 ans et plus, adapatation des tranches d'âge
    filter(AGE >= 15) %>%
    mutate(TRAGE = fct_drop(TRAGE)) %>%
    mutate(TRAGE = fct_recode(TRAGE, "[15;24]" = "2", "[25;34]" = "3", "[35;44]" = "4", "[45;54]" = "5"
    # Simplification des niveaux de diplôme et modalité de référence = bac
    mutate(DIPL = fct_explicit_na(DIPL)) %>%
    mutate(DIPL = fct_collapse(DIPL, "En cours" = c("1", "8"), "Inf. au lycée" = c("0", "2", "3"), "Lyc
    mutate(DIPL = fct_relevel(DIPL, "Bac")) %>%
    # Adapation des lieux de travail
    mutate(TYPLT = fct_explicit_na(TYPLT)) %>%
    mutate(TYPLT = fct_recode(TYPLT, "Bureau" = "1", "Commerce, boutique, grandes surfaces" = "2", "Ent
    # Adapation de la variabilité des lieux de travail
    mutate(ULTRAV = fct_explicit_na(ULTRAV)) %>%
    mutate(ULTRAV = fct_recode(ULTRAV, "Unique, en dehors du domicile" = "1", "Unique, au domicile" = "
    # Adaptation de la localisation du lieu de travail
    mutate(LTRAVCOUR = fct_explicit_na(LTRAVCOUR)) %>%
    mutate(LTRAVCOUR = fct_recode(LTRAVCOUR, "Paris" = "1", "Petite couronne" = "2", "Grande Couronne" =
    # Adapatation du permis de conduire
    # mutate(PERMVP = fct_explicit_na(PERMVP)) %>% # Commenté car que 1 personne et ça complique la lec
    mutate(PERMVP = fct_recode(PERMVP, "Oui" = "1", "Non" = "2", "Conduite accompagnée ou leçons" = "3"
    # Adaptation de l'abonnement TC
    mutate(ABONTC = fct_explicit_na(ABONTC)) %>%
    mutate(ABONTC = fct_other(ABONTC, keep = "1", other_level = "Oui")) %>%
```

```
    mutate(ABONTC = fct_recode(ABONTC, "Non" = "1")) %>%
    # Modalité de référence pour la gêne dans les déplacements = oui
    mutate(HANDI = fct_explicit_na(HANDI)) %>%
    mutate(HANDI = fct_relevel(HANDI, "2")) %>%
    mutate(HANDI = fct_recode(HANDI, "Oui" = "1", "Non" = "2"))
  data
}

individus_for_regression <- prepare_individus_for_regression(individus)

individus_pere_mere_for_regression <- individus_for_regression %>%
  # On ne garde que les pères et mères
  mutate(LIENPREF = fct_collapse(LIENPREF, "pere-mere" = c("1", "2"))) %>%
  mutate(LIENPREF = fct_other(LIENPREF, keep = "pere-mere")) %>%
  filter(LIENPREF == "pere-mere")
```

Régression

```
weighted_individus_pere_mere_for_regression <- svydesign(ids = ~1,
  data = individus_pere_mere_for_regression, weights = ~POIDSP)

reg_pere_mere <- svyglm(is_PCS56 ~ SEXE + TRAGE + DIPL + TYPLT + ULTRAV + LTRAVCOUR + PERMVP + ABONTC +
  weighted_individus_pere_mere_for_regression, family = quasibinomial())

weighted_individus_for_regression <- svydesign(ids = ~1,
  data = individus_for_regression, weights = ~POIDSP)

reg <- svyglm(is_PCS56 ~ SEXE + TRAGE + DIPL + TYPLT + ULTRAV + LTRAVCOUR + PERMVP + ABONTC + HANDI,
  weighted_individus_for_regression, family = quasibinomial())
```

Tous les individus

```
tbl_regression(reg, exponentiate = TRUE)
```

```
## Table printed with `knitr::kable()`, not {gt}. Learn why at
## http://www.danieldsjoberg.com/gtsummary/articles/rmarkdown.html
## To suppress this message, include `message = FALSE` in code chunk header.
```

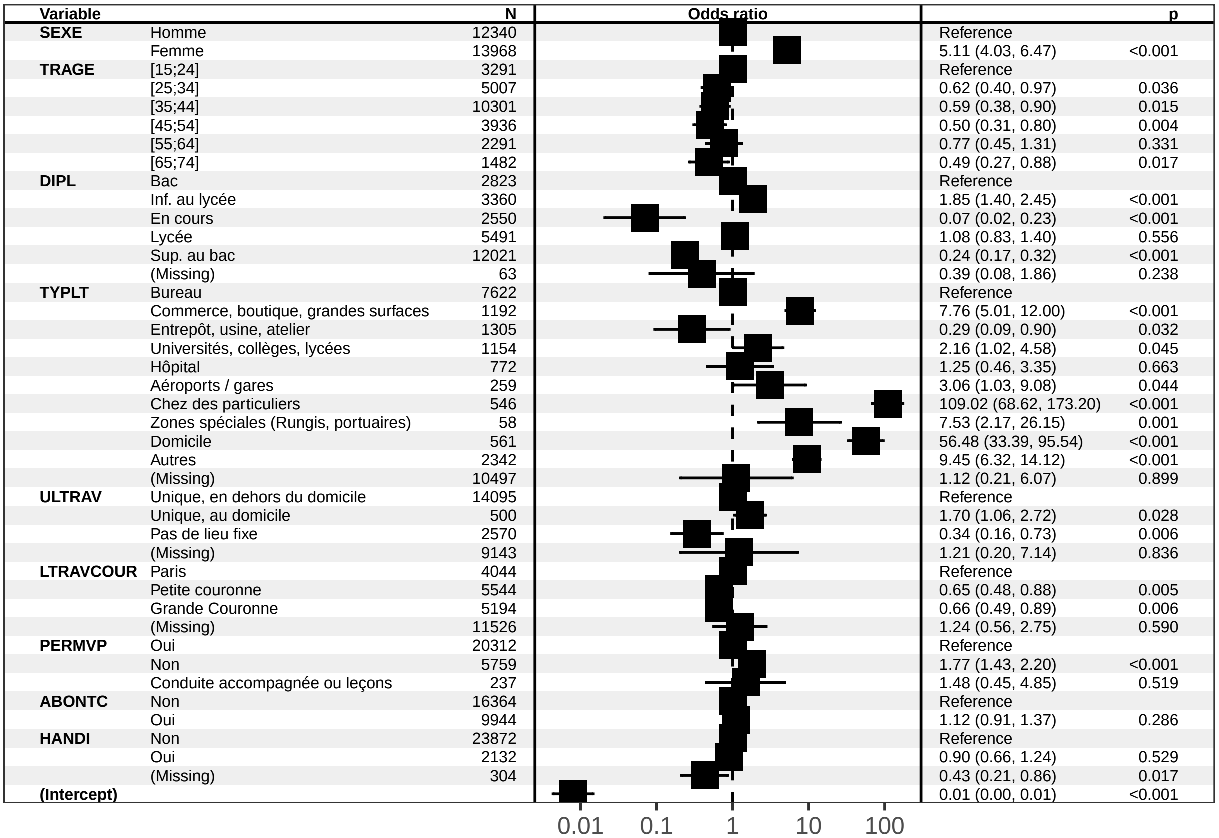
Characteristic	exp(Beta)	95% CI	p-value
SEXE			
Homme			
Femme	5.11	4.03, 6.47	<0.001
TRAGE			
[15;24]			
[25;34]	0.62	0.40, 0.97	0.036
[35;44]	0.59	0.38, 0.90	0.015
[45;54]	0.50	0.31, 0.80	0.004
[55;64]	0.77	0.45, 1.31	0.3
[65;74]	0.49	0.27, 0.88	0.017
DIPL			
Bac			
Inf. au lycée	1.85	1.40, 2.45	<0.001

Characteristic	exp(Beta)	95% CI	p-value
En cours	0.07	0.02, 0.23	<0.001
Lycée	1.08	0.83, 1.40	0.6
Sup. au bac	0.24	0.17, 0.32	<0.001
(Missing)	0.39	0.08, 1.86	0.2
TYPLT			
Bureau			
Commerce, boutique, grandes surfaces	7.76	5.01, 12.0	<0.001
Entrepôt, usine, atelier	0.29	0.09, 0.90	0.032
Universités, collèges, lycées	2.16	1.02, 4.58	0.045
Hôpital	1.25	0.46, 3.35	0.7
Aéroports / gares	3.06	1.03, 9.08	0.044
Chez des particuliers	109	68.6, 173	<0.001
Zones spéciales (Rungis, portuaires)	7.53	2.17, 26.1	0.001
Domicile	56.5	33.4, 95.5	<0.001
Autres	9.45	6.32, 14.1	<0.001
(Missing)	1.12	0.21, 6.07	0.9
ULTRAV			
Unique, en dehors du domicile			
Unique, au domicile	1.70	1.06, 2.72	0.028
Pas de lieu fixe	0.34	0.16, 0.73	0.006
(Missing)	1.21	0.20, 7.14	0.8
LTRAVCOUR			
Paris			
Petite couronne	0.65	0.48, 0.88	0.005
Grande Couronne	0.66	0.49, 0.89	0.006
(Missing)	1.24	0.56, 2.75	0.6
PERMVP			
Oui			
Non	1.77	1.43, 2.20	<0.001
Conduite accompagnée ou leçons	1.48	0.45, 4.85	0.5
ABONTC			
Non			
Oui	1.12	0.91, 1.37	0.3
HANDI			
Non			
Oui	0.90	0.66, 1.24	0.5
(Missing)	0.43	0.21, 0.86	0.017

```r
forest_model(reg)
```

```
## Warning: `data_frame()` is deprecated as of tibble 1.1.0.
## Please use `tibble()` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_warnings()` to see where this warning was generated.

## Warning: Ignoring unknown aesthetics: x
```



Variable		N	Odds ratio		p
SEXE	Homme	12340		Reference	
	Femme	13968		5.11 (4.03, 6.47)	<0.001
TRAGE	[15;24]	3291		Reference	
	[25;34]	5007		0.62 (0.40, 0.97)	0.036
	[35;44]	10301		0.59 (0.38, 0.90)	0.015
	[45;54]	3936		0.50 (0.31, 0.80)	0.004
	[55;64]	2291		0.77 (0.45, 1.31)	0.331
	[65;74]	1482		0.49 (0.27, 0.88)	0.017
DIPL	Bac	2823		Reference	
	Inf. au lycée	3360		1.85 (1.40, 2.45)	<0.001
	En cours	2550		0.07 (0.02, 0.23)	<0.001
	Lycée	5491		1.08 (0.83, 1.40)	0.556
	Sup. au bac	12021		0.24 (0.17, 0.32)	<0.001
	(Missing)	63		0.39 (0.08, 1.86)	0.238
TYPLT	Bureau	7622		Reference	
	Commerce, boutique, grandes surfaces	1192		7.76 (5.01, 12.00)	<0.001
	Entrepôt, usine, atelier	1305		0.29 (0.09, 0.90)	0.032
	Universités, collèges, lycées	1154		2.16 (1.02, 4.58)	0.045
	Hôpital	772		1.25 (0.46, 3.35)	0.663
	Aéroports / gares	259		3.06 (1.03, 9.08)	0.044
	Chez des particuliers	546		109.02 (68.62, 173.20)	<0.001
	Zones spéciales (Rungis, portuaires)	58		7.53 (2.17, 26.15)	0.001
	Domicile	561		56.48 (33.39, 95.54)	<0.001
	Autres	2342		9.45 (6.32, 14.12)	<0.001
	(Missing)	10497		1.12 (0.21, 6.07)	0.899
ULTRAV	Unique, en dehors du domicile	14095		Reference	
	Unique, au domicile	500		1.70 (1.06, 2.72)	0.028
	Pas de lieu fixe	2570		0.34 (0.16, 0.73)	0.006
	(Missing)	9143		1.21 (0.20, 7.14)	0.836
LTRAVCOUR	Paris	4044		Reference	
	Petite couronne	5544		0.65 (0.48, 0.88)	0.005
	Grande Couronne	5194		0.66 (0.49, 0.89)	0.006
	(Missing)	11526		1.24 (0.56, 2.75)	0.590
PERMVP	Oui	20312		Reference	
	Non	5759		1.77 (1.43, 2.20)	<0.001
	Conduite accompagnée ou leçons	237		1.48 (0.45, 4.85)	0.519
ABONTC	Non	16364		Reference	
	Oui	9944		1.12 (0.91, 1.37)	0.286
HANDI	Non	23872		Reference	
	Oui	2132		0.90 (0.66, 1.24)	0.529
	(Missing)	304		0.43 (0.21, 0.86)	0.017
(Intercept)				0.01 (0.00, 0.01)	<0.001

Que les pères et mères

```r
tbl_regression(reg_pere_mere, exponentiate = TRUE)
```

```
## Table printed with `knitr::kable()`, not {gt}. Learn why at
## http://www.danieldsjoberg.com/gtsummary/articles/rmarkdown.html
## To suppress this message, include `message = FALSE` in code chunk header.
```

Characteristic	exp(Beta)	95% CI	p-value
SEXE			
Homme			
Femme	5.06	3.93, 6.51	<0.001
TRAGE			
[15;24]			
[25;34]	0.58	0.31, 1.06	0.078
[35;44]	0.57	0.32, 1.03	0.062
[45;54]	0.49	0.26, 0.92	0.025
[55;64]	0.74	0.38, 1.44	0.4
[65;74]	0.45	0.22, 0.92	0.028
DIPL			
Bac			
Inf. au lycée	2.05	1.53, 2.76	<0.001
En cours	0.38	0.06, 2.45	0.3
Lycée	1.16	0.88, 1.52	0.3
Sup. au bac	0.25	0.18, 0.35	<0.001
(Missing)	0.53	0.11, 2.64	0.4

Characteristic	exp(Beta)	95% CI	p-value
TYPLT			
Bureau			
Commerce, boutique, grandes surfaces	7.18	4.51, 11.4	<0.001
Entrepôt, usine, atelier	0.30	0.10, 0.94	0.038
Universités, collèges, lycées	2.00	0.90, 4.43	0.088
Hôpital	1.20	0.42, 3.42	0.7
Aéroports / gares	2.52	0.74, 8.58	0.14
Chez des particuliers	111	69.3, 179	<0.001
Zones spéciales (Rungis, portuaires)	7.66	2.17, 27.0	0.002
Domicile	57.5	33.4, 98.8	<0.001
Autres	8.99	5.94, 13.6	<0.001
(Missing)	0.86	0.08, 9.72	>0.9
ULTRAV			
Unique, en dehors du domicile			
Unique, au domicile	1.72	1.06, 2.80	0.028
Pas de lieu fixe	0.31	0.14, 0.69	0.004
(Missing)	1.50	0.12, 18.2	0.7
LTRAVCOUR			
Paris			
Petite couronne	0.71	0.52, 0.97	0.031
Grande Couronne	0.68	0.50, 0.92	0.012
(Missing)	1.35	0.59, 3.07	0.5
PERMVP			
Oui			
Non	1.99	1.61, 2.48	<0.001
Conduite accompagnée ou leçons	1.30	0.29, 5.90	0.7
ABONTC			
Non			
Oui	1.13	0.91, 1.40	0.3
HANDI			
Non			
Oui	0.97	0.70, 1.35	0.9
(Missing)	0.47	0.21, 1.02	0.055

```r
forest_model(reg_pere_mere)
```

```
## Warning: Ignoring unknown aesthetics: x
```

Variable		N	Odds ratio	p
SEXE	Homme	10590	Reference	
	Femme	12303	5.06 (3.93, 6.51)	<0.001
TRAGE	[15;24]	643	Reference	
	[25;34]	4543	0.58 (0.31, 1.06)	0.078
	[35;44]	10137	0.57 (0.32, 1.03)	0.062
	[45;54]	3880	0.49 (0.26, 0.92)	0.025
	[55;64]	2255	0.74 (0.38, 1.44)	0.376
	[65;74]	1435	0.45 (0.22, 0.92)	0.028
DIPL	Bac	2605	Reference	
	Inf. au lycée	3124	2.05 (1.53, 2.76)	<0.001
	En cours	449	0.38 (0.06, 2.45)	0.311
	Lycée	5089	1.16 (0.88, 1.52)	0.295
	Sup. au bac	11571	0.25 (0.18, 0.35)	<0.001
	(Missing)	55	0.53 (0.11, 2.64)	0.439
TYPLT	Bureau	7371	Reference	
	Commerce, boutique, grandes surfaces	1033	7.18 (4.51, 11.41)	<0.001
	Entrepôt, usine, atelier	1209	0.30 (0.10, 0.94)	0.038
	Universités, collèges, lycées	1127	2.00 (0.90, 4.43)	0.088
	Hôpital	741	1.20 (0.42, 3.42)	0.734
	Aéroports / gares	243	2.52 (0.74, 8.58)	0.139
	Chez des particuliers	510	111.32 (69.27, 178.90)	<0.001
	Zones spéciales (Rungis, portuaires)	54	7.66 (2.17, 26.99)	0.002
	Domicile	553	57.46 (33.43, 98.75)	<0.001
	Autres	2199	8.99 (5.94, 13.61)	<0.001
	(Missing)	7853	0.86 (0.08, 9.72)	0.903
ULTRAV	Unique, en dehors du domicile	12623	Reference	
	Unique, au domicile	478	1.72 (1.06, 2.80)	0.028
	Pas de lieu fixe	2334	0.31 (0.14, 0.69)	0.004
	(Missing)	7458	1.50 (0.12, 18.19)	0.750
LTRAVCOUR	Paris	3556	Reference	
	Petite couronne	5002	0.71 (0.52, 0.97)	0.031
	Grande Couronne	4714	0.68 (0.50, 0.92)	0.012
	(Missing)	9621	1.35 (0.59, 3.07)	0.479
PERMVP	Oui	19178	Reference	
	Non	3649	1.99 (1.61, 2.48)	<0.001
	Conduite accompagnée ou leçons	66	1.30 (0.29, 5.90)	0.735
ABONTC	Non	15109	Reference	
	Oui	7784	1.13 (0.91, 1.40)	0.280
HANDI	Non	20758	Reference	
	Oui	1947	0.97 (0.70, 1.35)	0.868
	(Missing)	188	0.47 (0.21, 1.02)	0.055
(Intercept)			0.01 (0.00, 0.02)	<0.001